

SSD: Single Shot MultiBox Detector

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Abstract. We present a method for detecting objects in images using a single deep neural network. Our approach, named SSD, discretizes the output space of bounding boxes into a set of default boxes over different aspect ratios and scales per feature map location. At prediction time, the network generates scores for the presence of each object category in each default box and produces adjustments to the box to better match the object shape. Additionally, the network combines predictions from multiple feature maps with different resolutions to naturally handle objects of various sizes. SSD is simple relative to methods that require object proposals because it completely eliminates proposal generation and subsequent pixel or feature resampling stages and encapsulates all computation in a single network. This makes SSD easy to train and straightforward to integrate into systems that require a detection component. Experimental results on the PASCAL VOC, COCO, and ILSVRC datasets confirm that SSD has competitive accuracy to methods that utilize an additional object proposal step and is much faster, while providing a unified framework for both training and inference. For 300×300 input, SSD achieves 74.3% mAP¹ on VOC2007 test at 59 FPS on a Nvidia Titan X and for 512×512 input, SSD achieves 76.9% mAP, outperforming a comparable state-of-the-art Faster R-CNN model. Compared to other single stage methods, SSD has much better accuracy even with a smaller input image size. Code is available at: <https://github.com/weiliu89/caffe/tree/ssd>.

Keywords: Real-time Object Detection; Convolutional Neural Network

1 Introduction

Current state-of-the-art object detection systems are variants of the following approach: hypothesize bounding boxes, resample pixels or features for each box, and apply a high-quality classifier. This pipeline has prevailed on detection benchmarks since the Selective Search work [1] through the current leading results on PASCAL VOC, COCO, and ILSVRC detection all based on Faster R-CNN[2] albeit with deeper features such as [3]. While accurate, these approaches have been too computationally intensive for embedded systems and, even with high-end hardware, too slow for real-time applications.

¹ We achieved even better results using an improved data augmentation scheme in follow-on experiments: 77.2% mAP for 300×300 input and 79.8% mAP for 512×512 input on VOC2007. Please see Sec. 3.6 for details.

SSD: 单次多框检测器

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摘要。我们提出了一种使用单一深度神经网络检测图像中物体的方法。我们的方法名为SSD，将边界框的输出空间离散化为每个特征图位置上不同宽高比和尺度的一组默认框。在预测时，网络会为每个默认框中存在的每个物体类别生成分数，并对框进行调整以更好地匹配物体形状。此外，网络结合了来自不同分辨率的多层特征图的预测，从而自然地处理各种尺寸的物体。相对于需要物体提议的方法，SSD较为简单，因为它完全消除了提议生成及后续的像素或特征重采样阶段，并将所有计算封装在单一网络中。这使得SSD易于训练，并能直接集成到需要检测组件的系统中。在PASCAL VOC、COCO和ILSVRC数据集上的实验结果证实，SSD与那些使用额外物体提议步骤的方法相比具有相当的准确性，且速度更快，同时为训练和推理提供了一个统一的框架。对于 300×300 的输入，SSD在VOC2007测试集上达到74.3%的mAP¹，在Nvidia Titan X上以59 FPS运行；对于 512×512 的输入，SSD达到76.9%的mAP，优于同类最先进的Faster R-CNN模型。与其他单阶段方法相比，即使输入图像尺寸较小，SSD仍具有更高的准确性。代码发布于：<https://github.com/weiliu89/caffe/tree/ssd>。

关键词：实时目标检测；卷积神经网络

1 引言

当前最先进的物体检测系统是以下方法的变体：假设边界框，为每个框重新采样像素或特征，并应用高质量的分类器。自选择性搜索工作[1]以来，这一流程在检测基准测试中一直占据主导地位，直至当前在PASCAL VOC、COCO和ILSVRC检测中领先的结果均基于Faster R-CNN[2]，尽管采用了如[3]等更深层的特征。这些方法虽然精确，但对嵌入式系统而言计算量过大，即使使用高端硬件，对于实时应用来说也过于缓慢。

¹ We achieved even better results using an improved data augmentation scheme in follow-on experiments: 77.2% mAP for 300×300 input and 79.8% mAP for 512×512 input on VOC2007. Please see Sec. 3.6 for details.

Often detection speed for these approaches is measured in seconds per frame (SPF), and even the fastest high-accuracy detector, Faster R-CNN, operates at only 7 frames per second (FPS). There have been many attempts to build faster detectors by attacking each stage of the detection pipeline (see related work in Sec. 4), but so far, significantly increased speed comes only at the cost of significantly decreased detection accuracy.

This paper presents the first deep network based object detector that does not re-sample pixels or features for bounding box hypotheses *and* is as accurate as approaches that do. This results in a significant improvement in speed for high-accuracy detection (59 FPS with mAP 74.3% on VOC2007 *test*, vs. Faster R-CNN 7 FPS with mAP 73.2% or YOLO 45 FPS with mAP 63.4%). The fundamental improvement in speed comes from eliminating bounding box proposals and the subsequent pixel or feature resampling stage. We are not the first to do this (cf [4,5]), but by adding a series of improvements, we manage to increase the accuracy significantly over previous attempts. Our improvements include using a small convolutional filter to predict object categories and offsets in bounding box locations, using separate predictors (filters) for different aspect ratio detections, and applying these filters to multiple feature maps from the later stages of a network in order to perform detection at multiple scales. With these modifications—especially using multiple layers for prediction at different scales—we can achieve high-accuracy using relatively low resolution input, further increasing detection speed. While these contributions may seem small independently, we note that the resulting system improves accuracy on real-time detection for PASCAL VOC from 63.4% mAP for YOLO to 74.3% mAP for our SSD. This is a larger relative improvement in detection accuracy than that from the recent, very high-profile work on residual networks [3]. Furthermore, significantly improving the speed of high-quality detection can broaden the range of settings where computer vision is useful.

We summarize our contributions as follows:

- We introduce SSD, a single-shot detector for multiple categories that is faster than the previous state-of-the-art for single shot detectors (YOLO), and significantly more accurate, in fact as accurate as slower techniques that perform explicit region proposals and pooling (including Faster R-CNN).
- The core of SSD is predicting category scores and box offsets for a fixed set of default bounding boxes using small convolutional filters applied to feature maps.
- To achieve high detection accuracy we produce predictions of different scales from feature maps of different scales, and explicitly separate predictions by aspect ratio.
- These design features lead to simple end-to-end training and high accuracy, even on low resolution input images, further improving the speed vs accuracy trade-off.
- Experiments include timing and accuracy analysis on models with varying input size evaluated on PASCAL VOC, COCO, and ILSVRC and are compared to a range of recent state-of-the-art approaches.

2 The Single Shot Detector (SSD)

This section describes our proposed SSD framework for detection (Sec. 2.1) and the associated training methodology (Sec. 2.2). Afterwards, Sec. 3 presents dataset-specific model details and experimental results.

这些方法的检测速度通常以秒每帧（SPF）来衡量，即便是最快的高精度检测器 Faster R-CNN，其运行速度也仅为每秒7帧（FPS）。虽然已有许多研究尝试通过改进检测流程的每个环节来构建更快的检测器（参见第4节的相关工作），但迄今为止，速度的显著提升往往伴随着检测精度的大幅下降。

本文提出了首个基于深度网络的目标检测器，它无需为边界框假设~~and~~重新采样像素或特征，且准确度与采用重采样的方法相当。这为高精度检测带来了显著的速度提升（在VOC2007测试集上达到59 FPS和74.3% mAP，而Faster R-CNN为7 FPS/73.2% mAP，YOLO为45 FPS/63.4% mAP）。速度的根本性提升源于消除了边界框提议及后续的像素或特征重采样阶段。我们并非首次尝试该方法（参见[4,5]），但通过一系列改进，我们显著提升了先前尝试的准确度。我们的改进包括：使用小型卷积滤波器预测目标类别和边界框位置偏移；为不同宽高比的检测使用独立的预测器（滤波器）；将这些滤波器应用于网络后期阶段的多层特征图，以实现多尺度检测。通过这些改进——尤其是使用多层特征进行多尺度预测——我们能够以相对低分辨率的输入实现高精度，从而进一步提升检测速度。尽管这些改进各自看似微小，但我们注意到最终系统将PASCAL VOC的实时检测准确度从YOLO的63.4% mAP提升至我们SSD的74.3% mAP。这一检测准确度的相对提升幅度，甚至超过了近期备受关注的残差网络研究[3]。此外，高质量检测速度的显著提升，能够拓展计算机视觉技术的适用场景范围。

我们将我们的贡献总结如下：

- 我们提出了SSD，一种用于多类别的单次检测器，它比之前最先进的单次检测器（YOLO）更快，并且显著更准确，实际上与执行显式区域提议和池化的较慢技术（包括Faster R-CNN）一样准确。 SSD的核心是使用应用于特征图的小型卷积滤波器，为一组固定的默认边界框预测类别分数和边界框偏移。
- 为了实现高检测精度，我们从不同尺度的特征图中生成不同尺度的预测，并通过宽高比显式分离预测。 这些设计特点使得训练简单且端到端，即使是在低分辨率输入图像上也能实现高精度，进一步优化了速度与准确性的权衡。
- 实验包括在PASCAL VOC、COCO和ILSVRC上评估的不同输入尺寸模型的时序和准确性分析，并与一系列最新的先进方法进行了比较。

2 单次检测器（SSD）

本节介绍了我们提出的用于检测的SSD框架（第2.1节）以及相关的训练方法（第2.2节）。随后，第3节展示了针对特定数据集的模型细节和实验结果。

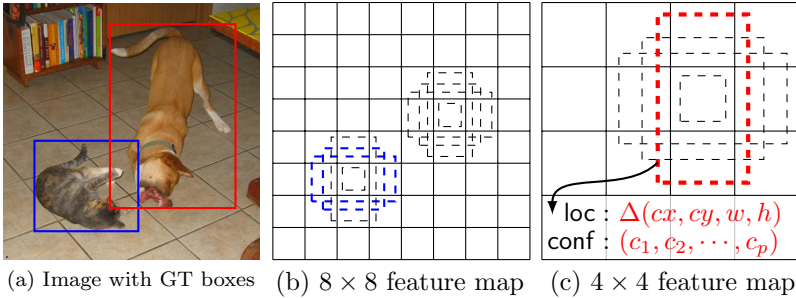


Fig. 1: SSD framework. (a) SSD only needs an input image and ground truth boxes for each object during training. In a convolutional fashion, we evaluate a small set (e.g. 4) of default boxes of different aspect ratios at each location in several feature maps with different scales (e.g. 8×8 and 4×4 in (b) and (c)). For each default box, we predict both the shape offsets and the confidences for all object categories ($(c_1, c_2, \dots, c_p)$). At training time, we first match these default boxes to the ground truth boxes. For example, we have matched two default boxes with the cat and one with the dog, which are treated as positives and the rest as negatives. The model loss is a weighted sum between localization loss (e.g. Smooth L1 [6]) and confidence loss (e.g. Softmax).

2.1 Model

The SSD approach is based on a feed-forward convolutional network that produces a fixed-size collection of bounding boxes and scores for the presence of object class instances in those boxes, followed by a non-maximum suppression step to produce the final detections. The early network layers are based on a standard architecture used for high quality image classification (truncated before any classification layers), which we will call the base network². We then add auxiliary structure to the network to produce detections with the following key features:

Multi-scale feature maps for detection We add convolutional feature layers to the end of the truncated base network. These layers decrease in size progressively and allow predictions of detections at multiple scales. The convolutional model for predicting detections is different for each feature layer (*cf* Overfeat[4] and YOLO[5] that operate on a single scale feature map).

Convolutional predictors for detection Each added feature layer (or optionally an existing feature layer from the base network) can produce a fixed set of detection predictions using a set of convolutional filters. These are indicated on top of the SSD network architecture in Fig. 2. For a feature layer of size $m \times n$ with p channels, the basic element for predicting parameters of a potential detection is a $3 \times 3 \times p$ *small kernel* that produces either a score for a category, or a shape offset relative to the default box coordinates. At each of the $m \times n$ locations where the kernel is applied, it produces an output value. The bounding box offset output values are measured relative to a default

² We use the VGG-16 network as a base, but other networks should also produce good results.

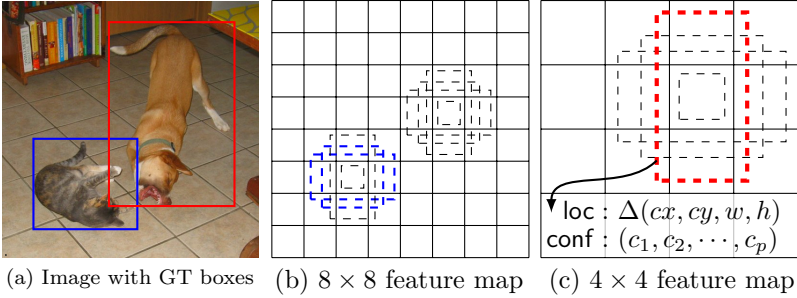


图1: SSD框架。(a) 训练时SSD仅需输入图像及每个物体的真实边界框。我们以卷积方式在多个不同尺度的特征图(如(b)和(c)中 8×8 与 4×4 的示例)的每个位置处, 评估一组少量(如4个)不同宽高比的默认框。针对每个默认框, 我们同时预测所有物体类别的形状偏移量及置信度 $(c_1, c_2, \dots, c_p)$ 。训练时, 我们首先将这些默认框与真实边界框进行匹配。例如, 图中两个默认框与猫匹配成功, 一个默认框与狗匹配成功, 这些匹配成功的框被视为正样本, 其余则为负样本。模型损失由定位损失(如Smooth L1 [6])与置信度损失(如Softmax)加权求和构成。

2.1 模型

SSD方法基于一个前馈卷积网络, 该网络生成固定尺寸的边界框集合及对应框中目标类别实例存在的评分, 随后通过非极大值抑制步骤生成最终检测结果。早期网络层采用用于高质量图像分类的标准架构(在任意分类层之前截断), 我们将其称为基础网络²。随后我们在网络中增加辅助结构, 以实现以下关键特性的检测:

用于检测的多尺度特征图 我们在截断的基础网络末端添加了卷积特征层。这些层的尺寸逐渐减小, 使得能够在多个尺度上进行检测预测。每个特征层的检测预测卷积模型各不相同 (*cf* Overfeat[4] 和 YOLO[5] 则是在单一尺度的特征图上进行操作)。

用于检测的卷积预测器 每个新增的特征层(或可选地, 从基础网络中选取的现有特征层)可以使用一组卷积滤波器生成固定的检测预测集。这些在图2的SSD网络架构顶部已标明。对于一个尺寸为 $m \times n$ 、通道数为 p 的特征层, 预测潜在检测参数的基本单元是一个 $3 \times 3 \times p$ *small kernel* 的卷积核, 它会为每个类别生成一个分数, 或生成相对于默认框坐标的形状偏移。在应用卷积核的 $m \times n$ 个位置中的每一个, 它都会产生一个输出值。边界框偏移的输出值是相对于默认框坐标进行度量的。

² We use the VGG-16 network as a base, but other networks should also produce good results.

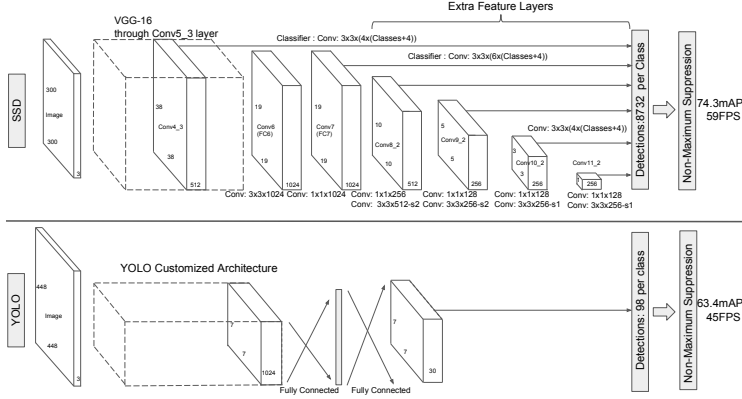


Fig. 2: A comparison between two single shot detection models: SSD and YOLO [5]. Our SSD model adds several feature layers to the end of a base network, which predict the offsets to default boxes of different scales and aspect ratios and their associated confidences. SSD with a 300×300 input size significantly outperforms its 448×448 YOLO counterpart in accuracy on VOC2007 test while also improving the speed.

box position relative to each feature map location (*cf* the architecture of YOLO[5] that uses an intermediate fully connected layer instead of a convolutional filter for this step).

Default boxes and aspect ratios We associate a set of default bounding boxes with each feature map cell, for multiple feature maps at the top of the network. The default boxes tile the feature map in a convolutional manner, so that the position of each box relative to its corresponding cell is fixed. At each feature map cell, we predict the offsets relative to the default box shapes in the cell, as well as the per-class scores that indicate the presence of a class instance in each of those boxes. Specifically, for each box out of k at a given location, we compute c class scores and the 4 offsets relative to the original default box shape. This results in a total of $(c + 4)k$ filters that are applied around each location in the feature map, yielding $(c + 4)kmn$ outputs for a $m \times n$ feature map. For an illustration of default boxes, please refer to Fig. 1. Our default boxes are similar to the *anchor boxes* used in Faster R-CNN [2], however we apply them to several feature maps of different resolutions. Allowing different default box shapes in several feature maps let us efficiently discretize the space of possible output box shapes.

2.2 Training

The key difference between training SSD and training a typical detector that uses region proposals, is that ground truth information needs to be assigned to specific outputs in the fixed set of detector outputs. Some version of this is also required for training in YOLO[5] and for the region proposal stage of Faster R-CNN[2] and MultiBox[7]. Once this assignment is determined, the loss function and back propagation are applied end-to-end. Training also involves choosing the set of default boxes and scales for detection as well as the hard negative mining and data augmentation strategies.

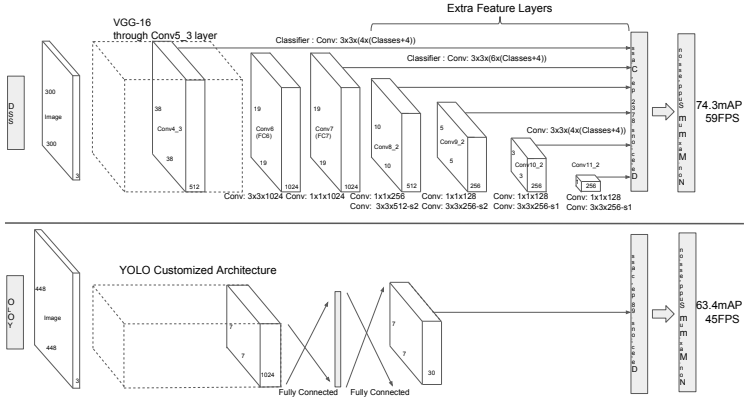


图2：两种单次检测模型SSD与YOLO的对比[5]。我们的SSD模型在基础网络末端添加了若干特征层，用于预测不同尺度与宽高比的默认边界框偏移量及其对应置信度。采用300×300输入尺寸的SSD在VOC2007测试集上的精度显著超越其448×448尺寸的YOLO对比模型，同时实现了速度提升。

相对于每个特征图位置的框位置（ $\{v^*\}$ YOLO[5]的架构在此步骤中使用了一个中间全连接层，而不是卷积滤波器）。

默认框与宽高比 我们为网络顶部的多个特征图中的每个特征图单元关联一组默认边界框。默认框以卷积方式平铺特征图，因此每个框相对于其对应单元的位置是固定的。在每个特征图单元中，我们预测相对于单元内默认框形状的偏移量，以及表示每个框中是否存在类别实例的各类别得分。具体而言，对于给定位置处的 k 个框中的每一个，我们计算 c 个类别得分和相对于原始默认框形状的4个偏移量。这将在特征图中每个位置周围应用总计 $(c + 4)k$ 个滤波器，从而为 $m \times n$ 特征图生成 $(c + 4)kmn$ 个输出。关于默认框的图示，请参见图1。我们的默认框类似于Faster R-CNN[2]中使用的*anchor boxes*，但我们将它们应用于多个不同分辨率的特征图。在多个特征图中允许不同的默认框形状，使我们能够高效离散化可能输出框形状的空间。

2.2 训练

SSD训练与使用区域建议的典型检测器训练的关键区别在于，需要将真实标注信息分配到固定检测器输出集合中的特定输出上。YOLO[5]的训练过程、Faster R-CNN[2]的区域建议阶段以及MultiBox[7]同样需要某种形式的此类分配。一旦确定这种分配关系，即可端到端地应用损失函数和反向传播。训练过程还包括选择用于检测的默认框集合与尺度范围，以及难负样本挖掘和数据增强策略。

Matching strategy During training we need to determine which default boxes correspond to a ground truth detection and train the network accordingly. For each ground truth box we are selecting from default boxes that vary over location, aspect ratio, and scale. We begin by matching each ground truth box to the default box with the best jaccard overlap (as in MultiBox [7]). Unlike MultiBox, we then match default boxes to any ground truth with jaccard overlap higher than a threshold (0.5). This simplifies the learning problem, allowing the network to predict high scores for multiple overlapping default boxes rather than requiring it to pick only the one with maximum overlap.

Training objective The SSD training objective is derived from the MultiBox objective [7,8] but is extended to handle multiple object categories. Let $x_{ij}^p = \{1, 0\}$ be an indicator for matching the i -th default box to the j -th ground truth box of category p . In the matching strategy above, we can have $\sum_i x_{ij}^p \geq 1$. The overall objective loss function is a weighted sum of the localization loss (loc) and the confidence loss (conf):

$$L(x, c, l, g) = \frac{1}{N} (L_{conf}(x, c) + \alpha L_{loc}(x, l, g)) \quad (1)$$

where N is the number of matched default boxes. If $N = 0$, we set the loss to 0. The localization loss is a Smooth L1 loss [6] between the predicted box (l) and the ground truth box (g) parameters. Similar to Faster R-CNN [2], we regress to offsets for the center (cx, cy) of the default bounding box (d) and for its width (w) and height (h).

$$\begin{aligned} L_{loc}(x, l, g) &= \sum_{i \in Pos} \sum_{m \in \{cx, cy, w, h\}} x_{ij}^k \text{smooth}_{L1}(l_i^m - \hat{g}_j^m) \\ \hat{g}_j^{cx} &= (g_j^{cx} - d_i^{cx}) / d_i^w & \hat{g}_j^{cy} &= (g_j^{cy} - d_i^{cy}) / d_i^h \\ \hat{g}_j^w &= \log\left(\frac{g_j^w}{d_i^w}\right) & \hat{g}_j^h &= \log\left(\frac{g_j^h}{d_i^h}\right) \end{aligned} \quad (2)$$

The confidence loss is the softmax loss over multiple classes confidences (c).

$$L_{conf}(x, c) = - \sum_{i \in Pos} x_{ij}^p \log(\hat{c}_i^p) - \sum_{i \in Neg} \log(\hat{c}_i^0) \quad \text{where} \quad \hat{c}_i^p = \frac{\exp(c_i^p)}{\sum_p \exp(c_i^p)} \quad (3)$$

and the weight term α is set to 1 by cross validation.

Choosing scales and aspect ratios for default boxes To handle different object scales, some methods [4,9] suggest processing the image at different sizes and combining the results afterwards. However, by utilizing feature maps from several different layers in a single network for prediction we can mimic the same effect, while also sharing parameters across all object scales. Previous works [10,11] have shown that using feature maps from the lower layers can improve semantic segmentation quality because the lower layers capture more fine details of the input objects. Similarly, [12] showed that adding global context pooled from a feature map can help smooth the segmentation results.

匹配策略 在训练过程中，我们需要确定哪些默认框与真实检测框相对应，并据此训练网络。针对每个真实框，我们从不同位置、宽高比和尺度的默认框中进行选择。我们首先将每个真实框与具有最佳杰卡德重叠度的默认框进行匹配（如MultiBox [7]所示）。与MultiBox不同的是，我们随后会将杰卡德重叠度高于阈值（0.5）的默认框与任意真实框进行匹配。这简化了学习问题，使网络能够为多个重叠的默认框预测高分，而不必要求它仅选择重叠度最高的那个。

训练目标 SSD的训练目标源自MultiBox目标[7,8]，但扩展至处理多目标类别。令 $x_{ij}^p = \{1,0\}$ 为指示函数，表示第 i 个默认框与第 j 个类别为 p 的真实框的匹配情况。根据上述匹配策略，可得 $\sum_i x_{ij}^p \geq 1$ 。整体目标损失函数由定位损失（loc）与置信度损失（conf）加权求和而得：

$$L(x, c, l, g) = \frac{1}{N} (L_{conf}(x, c) + \alpha L_{loc}(x, l, g)) \quad (1)$$

其中 N 是匹配的默认框的数量。如果 $N = 0$ ，我们将损失设为0。定位损失是预测框(l)与真实框(g)参数之间的平滑L1损失[6]。与Faster R-CNN[2]类似，我们对默认边界框(d)的中心(cx, cy)及其宽度(w)和高度(h)的偏移量进行回归。

$$\begin{aligned} L_{loc}(x, l, g) &= \sum_{i \in Pos} \sum_{m \in \{cx, cy, w, h\}} x_{ij}^k \text{smooth}_{L1}(l_i^m - \hat{g}_j^m) \\ \hat{g}_j^{cx} &= (g_j^{cx} - d_i^{cx}) / d_i^w & \hat{g}_j^{cy} &= (g_j^{cy} - d_i^{cy}) / d_i^h \\ \hat{g}_j^w &= \log\left(\frac{g_j^w}{d_i^w}\right) & \hat{g}_j^h &= \log\left(\frac{g_j^h}{d_i^h}\right) \end{aligned} \quad (2)$$

置信度损失是在多个类别置信度（ $\{v^*\}$ ）上的softmax损失。

$$L_{conf}(x, c) = - \sum_{i \in Pos} x_{ij}^p \log(\hat{c}_i^p) - \sum_{i \in Neg} \log(\hat{c}_i^0) \quad \text{where} \quad \hat{c}_i^p = \frac{\exp(c_i^p)}{\sum_p \exp(c_i^p)} \quad (3)$$

并且权重项 α 通过交叉验证被设为1。

为默认框选择尺度和长宽比 为了处理不同物体的尺度，一些方法[9]建议在不同尺寸下处理图像，然后将结果合并。然而，通过在单个网络中利用多个不同层的特征图进行预测，我们可以模拟相同的效果，同时还能在所有物体尺度间共享参数。先前的研究[10,11]表明，使用较低层的特征图可以提高语义分割的质量，因为较低层能捕捉输入物体更精细的细节。类似地，[12]的研究显示，从特征图中提取的全局上下文信息有助于平滑分割结果。

Motivated by these methods, we use both the lower and upper feature maps for detection. Figure 1 shows two exemplar feature maps (8×8 and 4×4) which are used in the framework. In practice, we can use many more with small computational overhead.

Feature maps from different levels within a network are known to have different (empirical) receptive field sizes [13]. Fortunately, within the SSD framework, the default boxes do not necessary need to correspond to the actual receptive fields of each layer. We design the tiling of default boxes so that specific feature maps learn to be responsive to particular scales of the objects. Suppose we want to use m feature maps for prediction. The scale of the default boxes for each feature map is computed as:

$$s_k = s_{\min} + \frac{s_{\max} - s_{\min}}{m - 1}(k - 1), \quad k \in [1, m] \quad (4)$$

where $s_{\min}$ is 0.2 and $s_{\max}$ is 0.9, meaning the lowest layer has a scale of 0.2 and the highest layer has a scale of 0.9, and all layers in between are regularly spaced. We impose different aspect ratios for the default boxes, and denote them as $a_r \in \{1, 2, 3, \frac{1}{2}, \frac{1}{3}\}$. We can compute the width ($w_k^a = s_k \sqrt{a_r}$) and height ($h_k^a = s_k / \sqrt{a_r}$) for each default box. For the aspect ratio of 1, we also add a default box whose scale is $s'_k = \sqrt{s_k s_{k+1}}$, resulting in 6 default boxes per feature map location. We set the center of each default box to $(\frac{i+0.5}{|f_k|}, \frac{j+0.5}{|f_k|})$, where $|f_k|$ is the size of the k -th square feature map, $i, j \in [0, |f_k|)$. In practice, one can also design a distribution of default boxes to best fit a specific dataset. How to design the optimal tiling is an open question as well.

By combining predictions for all default boxes with different scales and aspect ratios from all locations of many feature maps, we have a diverse set of predictions, covering various input object sizes and shapes. For example, in Fig. 1, the dog is matched to a default box in the 4×4 feature map, but not to any default boxes in the 8×8 feature map. This is because those boxes have different scales and do not match the dog box, and therefore are considered as negatives during training.

Hard negative mining After the matching step, most of the default boxes are negatives, especially when the number of possible default boxes is large. This introduces a significant imbalance between the positive and negative training examples. Instead of using all the negative examples, we sort them using the highest confidence loss for each default box and pick the top ones so that the ratio between the negatives and positives is at most 3:1. We found that this leads to faster optimization and a more stable training.

Data augmentation To make the model more robust to various input object sizes and shapes, each training image is randomly sampled by one of the following options:

- Use the entire original input image.
- Sample a patch so that the *minimum* jaccard overlap with the objects is 0.1, 0.3, 0.5, 0.7, or 0.9.
- Randomly sample a patch.

The size of each sampled patch is $[0.1, 1]$ of the original image size, and the aspect ratio is between $\frac{1}{2}$ and 2. We keep the overlapped part of the ground truth box if the center of it is in the sampled patch. After the aforementioned sampling step, each sampled patch is resized to fixed size and is horizontally flipped with probability of 0.5, in addition to applying some photo-metric distortions similar to those described in [14].

受这些方法的启发，我们同时使用低层和高层特征图进行检测。图1展示了框架中使用的两个示例特征图（ 8×8 和 4×4 ）。在实际应用中，我们可以使用更多特征图，而计算开销仅小幅增加。

已知网络内不同层次的特征图具有不同的（经验）感受野大小[13]。幸运的是，在SSD框架中，默认框并不必须与每一层的实际感受野相对应。我们设计默认框的平铺方式，使得特定特征图学会对特定尺度的物体产生响应。假设我们想使用 m 个特征图进行预测。每个特征图的默认框尺度计算如下：

$$s_k = s_{\min} + \frac{s_{\max} - s_{\min}}{m - 1}(k - 1), \quad k \in [1, m] \quad (4)$$

其中 $s_{\min}$ 为0.2， $s_{\max}$ 为0.9，意味着最底层尺度为0.2，最高层尺度为0.9，中间各层尺度均匀分布。我们为默认框设定了不同的宽高比，并将其表示为 $a_r \in \{1, 2, 3, \frac{1}{2}, \frac{1}{3}\}$ 。可计算每个默认框的宽度($w_k^a = s_k \sqrt{a_r}$)与高度($h_k^a = s_k / \sqrt{a_r}$)。对于宽高比为1的情况，额外添加尺度为 $s'_k = \sqrt{s_k s_{k+1}}$ 的默认框，使得每个特征图位置对应6个默认框。各默认框的中心点设置为 $(\frac{i+0.5}{|f_k|}, \frac{j+0.5}{|f_k|})$ ，其中 $|f_k|$ 表示第 k 个方形特征图的尺寸， $i, j \in [0, |f_k|)$ 。实际应用中也可根据特定数据集设计默认框的分布方案，如何构建最优平铺策略仍是一个开放性问题。

通过结合来自多个特征图所有位置、不同尺度和长宽比的默认框的预测，我们得到了一组多样化的预测，覆盖了各种输入对象的大小和形状。例如，在图1中，狗被匹配到 4×4 特征图中的一个默认框，但未与 8×8 特征图中的任何默认框匹配。这是因为这些框具有不同的尺度，与狗的边界框不匹配，因此在训练过程中被视为负样本。

硬负样本挖掘 在匹配步骤之后，大多数默认框都是负样本，尤其是在可能的默认框数量较多时。这导致正负训练样本之间存在显著的不平衡。我们并非使用所有负样本，而是根据每个默认框的最高置信度损失对它们进行排序，并选择损失最高的样本，使得负样本与正样本的比例最多为3:1。我们发现这种方法能加速优化过程并使训练更稳定。

数据增强 为了使模型对各种输入对象的大小和形状更加鲁棒，每个训练图像都通过以下选项之一随机采样：

- 使用完整的原始输入图像。 采样一个图像块，使其与目标物体的 *minimum* 杰卡德重叠度为 0.1、0.3、0.5、0.7 或 0.9。 随机采样一个图像块。

每个采样图像块的大小是原始图像尺寸的 $[0.1, 1]$ 倍，宽高比介于 $\frac{1}{2}$ 和2之间。若真实标注框的中心位于采样图像块内，则保留其重叠部分。完成上述采样步骤后，每个采样图像块会被调整为固定尺寸，并以0.5的概率进行水平翻转，同时施加类似于[14]中所述的光度畸变处理。

3 Experimental Results

Base network Our experiments are all based on VGG16 [15], which is pre-trained on the ILSVRC CLS-LOC dataset [16]. Similar to DeepLab-LargeFOV [17], we convert fc6 and fc7 to convolutional layers, subsample parameters from fc6 and fc7, change pool5 from $2 \times 2 - s2$ to $3 \times 3 - s1$, and use the *à trous* algorithm [18] to fill the "holes". We remove all the dropout layers and the fc8 layer. We fine-tune the resulting model using SGD with initial learning rate 10^{-3} , 0.9 momentum, 0.0005 weight decay, and batch size 32. The learning rate decay policy is slightly different for each dataset, and we will describe details later. The full training and testing code is built on Caffe [19] and is open source at: <https://github.com/weiliu89/caffe/tree/ssd>.

3.1 PASCAL VOC2007

On this dataset, we compare against Fast R-CNN [6] and Faster R-CNN [2] on VOC2007 test (4952 images). All methods fine-tune on the same pre-trained VGG16 network.

Figure 2 shows the architecture details of the SSD300 model. We use conv4_3, conv7 (fc7), conv8_2, conv9_2, conv10_2, and conv11_2 to predict both location and confidences. We set default box with scale 0.1 on conv4_3³. We initialize the parameters for all the newly added convolutional layers with the "xavier" method [20]. For conv4_3, conv10_2 and conv11_2, we only associate 4 default boxes at each feature map location – omitting aspect ratios of $\frac{1}{3}$ and 3. For all other layers, we put 6 default boxes as described in Sec. 2.2. Since, as pointed out in [12], conv4_3 has a different feature scale compared to the other layers, we use the L2 normalization technique introduced in [12] to scale the feature norm at each location in the feature map to 20 and learn the scale during back propagation. We use the 10^{-3} learning rate for 40k iterations, then continue training for 10k iterations with 10^{-4} and 10^{-5} . When training on VOC2007 trainval, Table 1 shows that our low resolution SSD300 model is already more accurate than Fast R-CNN. When we train SSD on a larger 512×512 input image, it is even more accurate, surpassing Faster R-CNN by 1.7% mAP. If we train SSD with more (i.e. 07+12) data, we see that SSD300 is already better than Faster R-CNN by 1.1% and that SSD512 is 3.6% better. If we take models trained on COCO trainval35k as described in Sec. 3.4 and fine-tuning them on the 07+12 dataset with SSD512, we achieve the best results: 81.6% mAP.

To understand the performance of our two SSD models in more details, we used the detection analysis tool from [21]. Figure 3 shows that SSD can detect various object categories with high quality (large white area). The majority of its confident detections are correct. The recall is around 85-90%, and is much higher with "weak" (0.1 jaccard overlap) criteria. Compared to R-CNN [22], SSD has less localization error, indicating that SSD can localize objects better because it directly learns to regress the object shape and classify object categories instead of using two decoupled steps. However, SSD has more confusions with similar object categories (especially for animals), partly because we share locations for multiple categories. Figure 4 shows that SSD is very sensitive to the bounding box size. In other words, it has much worse performance on smaller

³ For SSD512 model, we add extra conv12_2 for prediction, set $s_{\min}$ to 0.15, and 0.07 on conv4_3.

3 实验结果

基础网络 我们的实验均基于VGG16 [15], 该模型已在ILSVRC CLS-LOC数据集 [16] 上进行了预训练。与DeepLab-LargeFOV [17] 类似, 我们将fc6和fc7转换为卷积层, 从fc6和fc7中下采样参数, 将pool5从 $2 \times 2 - s_2$ 改为 $3 \times 3 - s_1$, 并使用 *à trous* 算法 [18] 填充“空洞”。我们移除了所有的dropout层和fc8层。我们使用SGD对所得模型进行微调, 初始学习率为 10^{-3} , 动量为0.9, 权重衰减为0.0005, 批量大小为32。每个数据集的学习率衰减策略略有不同, 具体细节将在后文描述。完整的训练和测试代码基于Caffe [19] 构建, 并已在以下地址开源: <https://github.com/weiliu89/caffe/tree/ssd>。

3.1 PASCAL VOC2007

在该数据集上, 我们在VOC2007测试集 (4952张图像) 上对比了Fast R-CNN [6] 和Faster R-CNN [2] 的性能。所有方法均基于相同的预训练VGG16网络进行微调。图2展示了SSD300模型的架构细节。我们使用conv4_3、conv7 (fc7)、conv8_2、conv9_2、conv10_2和conv11_2层同时预测位置和置信度。在conv4_3³层上, 我们将默认框的尺度设置为0.1。所有新增卷积层的参数均采用“xavier”方法 [20] 进行初始化。对于conv4_3、conv10_2和conv11_2层, 我们在每个特征图位置仅关联4个默认框——省略了 $\frac{1}{3}$ 和3的长宽比。其余所有层均按照第2.2节所述设置6个默认框。由于文献 [12] 指出conv4_3层具有与其他层不同的特征尺度, 我们采用 [12] 中提出的L2归一化技术, 将特征图中每个位置的特征范数缩放至20, 并在反向传播过程中学习缩放因子。我们使用 10^{-3} 的学习率训练40k次迭代, 随后以 10^{-4} 和 10^{-5} 的学习率继续训练10k次迭代。在VOC2007 trainval数据集上训练时, 表1显示我们的低分辨率SSD300模型已比Fast R-CNN更精确。当使用更大的 512×512 输入图像训练SSD时, 其精度进一步提升, 以1.7%的mAP优势超越Faster R-CNN。若使用更多数据 (即07+12) 训练SSD, 可见SSD300已以1.1%的优势优于Faster R-CNN, 而SSD512则领先3.6%。若采用第3.4节所述在COCO trainval 35k上训练的模型, 并在07+12数据集上用SSD512进行微调, 我们获得了最佳结果: 81.6%的mAP。

为了更详细地了解我们两个SSD模型的性能, 我们使用了文献 [21] 中的检测分析工具。图3显示SSD能够高质量地检测多种物体类别 (图中白色区域面积较大)。其大部分高置信度检测结果都是正确的。召回率约为85-90%, 若采用“宽松”评判标准 (0.1雅卡尔重叠度) 则召回率更高。与R-CNN [22] 相比, SSD的定位误差更小, 这表明SSD能更精准地定位物体, 因为它直接学习回归物体形状并分类物体类别, 而非使用两个解耦的步骤。但SSD对相似物体类别 (尤其是动物) 更容易产生混淆, 部分原因是我们对多个类别共享了定位信息。图4表明SSD对边界框尺寸非常敏感, 换言之, 其在较小物体上的性能明显下降。

³ For SSD512 model, we add extra conv12_2 for prediction, set $s_{\min}$ to 0.15, and 0.07 on conv4_3.

Method	data	mAP	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv
Fast [6]	07	66.9	74.5	78.3	69.2	53.2	36.6	77.3	78.2	82.0	40.7	72.7	67.9	79.6	79.2	73.0	69.0	30.1	65.4	70.2	75.8	65.8
Fast [6]	07+12	70.0	77.0	78.1	69.3	59.4	38.3	81.6	78.6	86.7	42.8	78.8	68.9	84.7	82.0	76.6	69.9	31.8	70.1	74.8	80.4	70.4
Faster [2]	07	69.9	70.0	80.6	70.1	57.3	49.9	78.2	80.4	82.0	52.2	75.3	67.2	80.3	79.8	75.0	76.3	39.1	68.3	67.3	81.1	67.6
Faster [2]	07+12	73.2	76.5	79.0	70.9	65.5	52.1	83.1	84.7	86.4	52.0	81.9	65.7	84.8	84.6	77.5	76.7	38.8	73.6	73.9	83.0	72.6
Faster [2]	07+12+COCO	78.8	84.3	82.0	77.7	68.9	65.7	88.1	88.4	88.9	63.6	86.3	70.8	85.9	87.6	80.1	82.3	53.6	80.4	75.8	86.6	78.9
SSD300	07	68.0	73.4	77.5	64.1	59.0	38.9	75.2	80.8	78.5	46.0	67.8	69.2	76.6	82.1	77.0	72.5	41.2	64.2	69.1	78.0	68.5
SSD300	07+12	74.3	75.5	80.2	72.3	66.3	47.6	83.0	84.2	86.1	54.7	78.3	73.9	84.5	85.3	82.6	76.2	48.6	73.9	76.0	83.4	74.0
SSD300	07+12+COCO	79.6	80.9	86.3	79.0	76.2	57.6	87.3	88.2	88.6	60.5	85.4	76.7	87.5	89.2	84.5	81.4	55.0	81.9	81.5	85.9	78.9
SSD512	07	71.6	75.1	81.4	69.8	60.8	46.3	82.6	84.7	84.1	48.5	75.0	67.4	82.3	83.9	79.4	76.6	44.9	69.9	69.1	78.1	71.8
SSD512	07+12	76.8	82.4	84.7	78.4	73.8	53.2	86.2	87.5	86.0	57.8	83.1	70.2	84.9	85.2	83.9	79.7	50.3	77.9	73.9	82.5	75.3
SSD512	07+12+COCO	81.6	86.6	88.3	82.4	76.0	66.3	88.6	88.9	89.1	65.1	88.4	73.6	86.5	88.9	85.3	84.6	59.1	85.0	80.4	87.4	81.2

Table 1: **PASCAL VOC2007 test detection results.** Both Fast and Faster R-CNN use input images whose minimum dimension is 600. The two SSD models have exactly the same settings except that they have different input sizes (300×300 vs. 512×512). It is obvious that larger input size leads to better results, and more data always helps. Data: "07": VOC2007 `trainval`, "07+12": union of VOC2007 and VOC2012 `trainval`. "07+12+COCO": first train on COCO `trainval35k` then fine-tune on 07+12.

objects than bigger objects. This is not surprising because those small objects may not even have any information at the very top layers. Increasing the input size (e.g. from 300×300 to 512×512) can help improve detecting small objects, but there is still a lot of room to improve. On the positive side, we can clearly see that SSD performs really well on large objects. And it is very robust to different object aspect ratios because we use default boxes of various aspect ratios per feature map location.

3.2 Model analysis

To understand SSD better, we carried out controlled experiments to examine how each component affects performance. For all the experiments, we use the same settings and input size (300×300), except for specified changes to the settings or component(s).

	SSD300				
more data augmentation?		✓	✓	✓	✓
include $\{\frac{1}{2}, 2\}$ box?	✓		✓	✓	✓
include $\{\frac{1}{3}, 3\}$ box?	✓			✓	✓
use atrous?	✓	✓	✓		✓
VOC2007 test mAP	65.5	71.6	73.7	74.2	74.3

Table 2: **Effects of various design choices and components on SSD performance.**

Data augmentation is crucial. Fast and Faster R-CNN use the original image and the horizontal flip to train. We use a more extensive sampling strategy, similar to YOLO [5]. Table 2 shows that we can improve 8.8% mAP with this sampling strategy. We do not know how much our sampling strategy will benefit Fast and Faster R-CNN, but they are likely to benefit less because they use a feature pooling step during classification that is relatively robust to object translation by design.

方法 数据 mAP 航空器 自行车 鸟类 船只 瓶子 公交车 汽车 猫 椅子 牛 桌子 狗 马 摩托车 人 植物 羊 沙发 火车 电视 Fast [6] 07 66.9 74.5 78.3 69.2 53.2 36.6 77.3 78.2 82.0 40.7 72.7 67.9 79.6 79.2 73.0 69.0 30.1 65.4 70.2 75.8 65.8 Fast [6] 07+12 70.0 77.0 78.1 69.3 59.4 38.3 81.6 78.6 86.7 42.8 78.8 68.9 84.7 82.0 76.6 69.9 31.8 70.1 74.8 80.4 70.4 Faster [2] 07 69.9 70.0 80.6 70.1 57.3 49.9 78.2 80.4 82.0 52.2 75.3 67.2 80.3 79.8 75.0 76.3 39.1 68.3 67.3 81.1 67.6 Faster [2] 07+12 73.2 76.5 79.0 70.9 65.5 52.1 83.1 84.7 86.4 52.0 81.9 65.7 84.8 84.6 77.5 76.7 38.8 73.6 73.9 83.0 72.6 Faster [2] 07+12+COCO 78.8 84.3 82.0 77.7 68.9 65.7 88.1 88.4 88.9 63.6 86.3 70.8 85.9 87.6 80.1 82.3 53.6 80.4 75.8 86.6 78.4 SSD300 07 68.0 73.4 77.5 64.1 59.0 38.9 75.2 80.8 78.5 46.0 67.8 69.2 76.6 82.1 77.0 72.5 41.2 64.2 69.1 78.0 68.5 SSD300 07+12 7.4 3.7 5.0 76.2 57.6 87.3 88.2 88.6 60.5 85.4 76.7 87.5 89.2 84.5 81.4 55.0 81.9 81.5 85.9 78.9 SSD512 07 71.6 75.1 81.4 69.8 60.8 46.3 82.6 84.7 84.1 48.5 75.0 67.4 82.3 83.9 79.4 76.6 44.9 69.9 69.1 78.1 71.8 SSD512 07+12 76.8 82.4 84.7 78.4 73.8 53.2 86.2 87.5 86.0 57.8 83.1 70.2 84.9 85.2 83.9 79.7 50.3 77.9 73.9 82.5 75.3 SSD512 07+12+COCO 81.6 86.6 88.3 82.4 76.0 66.3 88.6 88.9 89.1 65.1 88.4 73.6 86.5 88.9 85.3 84.6 5.1 85.0 80.4 87.4 81.2

表1: PASCAL VOC2007测试集检测结果。Fast R-CNN与Faster R-CNN均采用最小边长为600的输入图像。两个SSD模型除输入尺寸不同外（300×300对比512×512）完全采用相同配置。显然更大的输入尺寸能带来更好的结果，而更多数据始终有益。数据标注说明: "07": VOC2007训练验证集, "07+12": VOC2007与VOC2012训练验证集的联合集。"07+12+COCO": 先在COCO trainval35k上训练，再在07+12上微调。

小物体比大物体更难检测。这并不令人意外，因为这些小物体可能在最高层根本没有信息。增加输入尺寸（例如从300×300提升至512×512）有助于改善小物体检测，但仍有很大改进空间。从积极的一面看，我们可以清楚地看到SSD在大物体上表现非常出色，并且对不同物体长宽比具有很强鲁棒性，因为我们在每个特征图位置使用了多种长宽比的默认框。

3.2 模型分析

为了更好地理解SSD，我们进行了对照实验，以检验每个组件如何影响性能。在所有实验中，除了对设置或组件进行特定更改外，我们使用相同的设置和输入尺寸（300 {v*} 300）。

	SSD300				
more data augmentation?		✓	✓	✓	✓
include { $\frac{1}{2}, 2$ } box?	✓			✓	✓
include { $\frac{1}{3}, 3$ } box?	✓			✓	✓
use atrous?	✓	✓	✓		✓
VOC2007 test mAP	65.5	71.6	73.7	74.2	74.3

表2: 各种设计选择和组件对SSD性能的影响。

数据增强至关重要。Fast R-CNN和Faster R-CNN使用原始图像和水平翻转进行训练。我们采用了更广泛的采样策略，类似于YOLO[5]。表2显示，通过这种采样策略，我们可以将mAP提升8.8%。我们尚不清楚我们的采样策略能为Fast R-CNN和Faster R-CNN带来多少增益，但它们的受益可能较小，因为它们在分类过程中使用了特征池化步骤，该步骤在设计上对目标平移具有相对较强的鲁棒性。

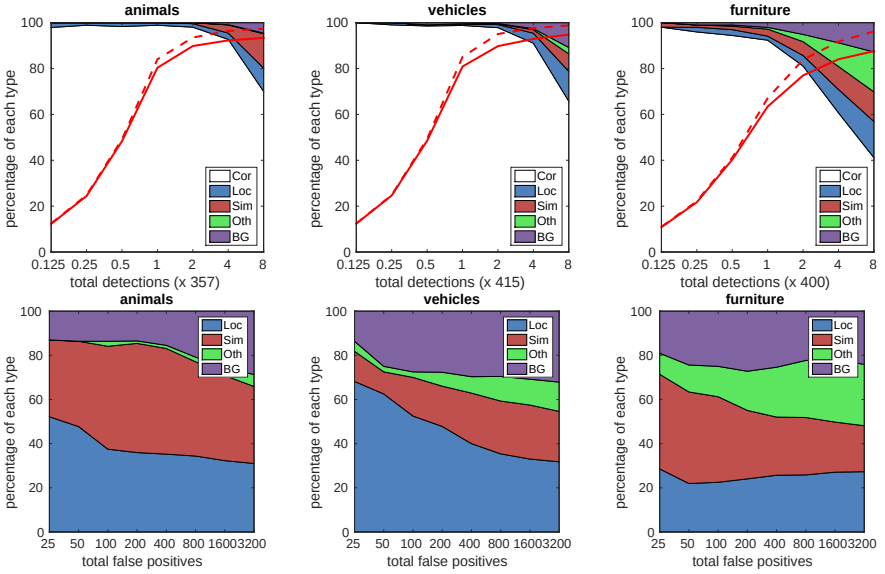


Fig. 3: **Visualization of performance for SSD512 on animals, vehicles, and furniture from VOC2007 test.** The top row shows the cumulative fraction of detections that are correct (Cor) or false positive due to poor localization (Loc), confusion with similar categories (Sim), with others (Oth), or with background (BG). The solid red line reflects the change of recall with strong criteria (0.5 jaccard overlap) as the number of detections increases. The dashed red line is using the weak criteria (0.1 jaccard overlap). The bottom row shows the distribution of top-ranked false positive types.

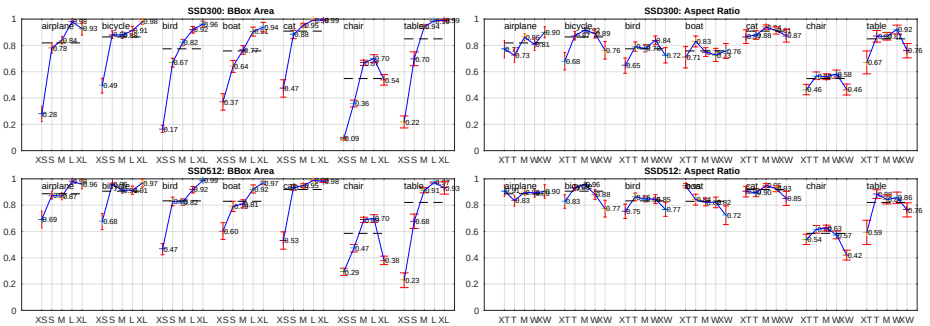


Fig. 4: **Sensitivity and impact of different object characteristics on VOC2007 test set using [21].** The plot on the left shows the effects of BBox Area per category, and the right plot shows the effect of Aspect Ratio. Key: BBox Area: XS=extra-small; S=small; M=medium; L=large; XL =extra-large. Aspect Ratio: XT=extra-tall/narrow; T=tall; M=medium; W=wide; XW =extra-wide.

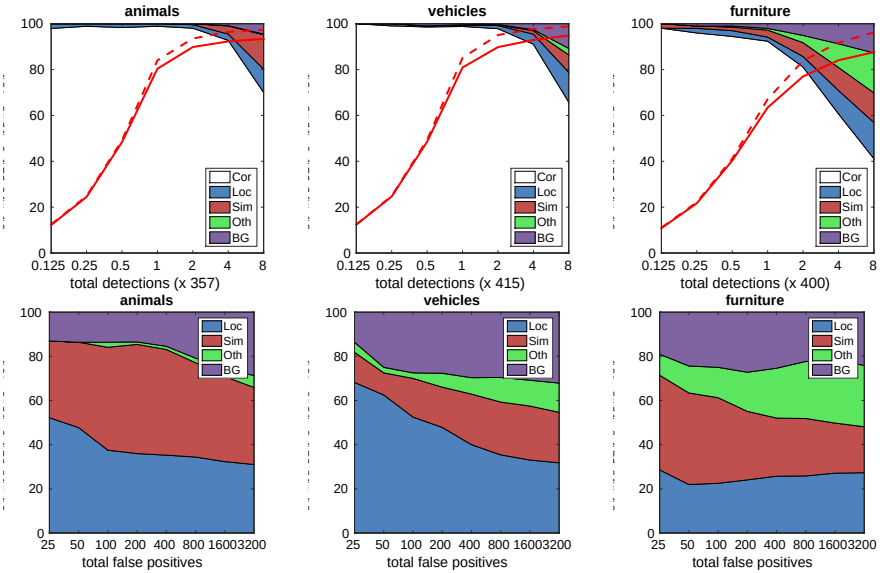


图3: SSD512在VOC2007测试集上对动物、车辆和家具类别的性能可视化。首行展示了检测结果中正确检测 (Cor) 或因定位偏差 (Loc)、与相似类别混淆 (Sim)、与其他类别混淆 (Oth) 以及与背景混淆 (BG) 导致的假阳性累积比例。红色实线反映了在严格标准 (0.5雅卡尔重叠度) 下, 召回率随检测数量增加的变化趋势; 红色虚线则对应宽松标准 (0.1雅卡尔重叠度)。底行显示了排名靠前的假阳性类型分布。

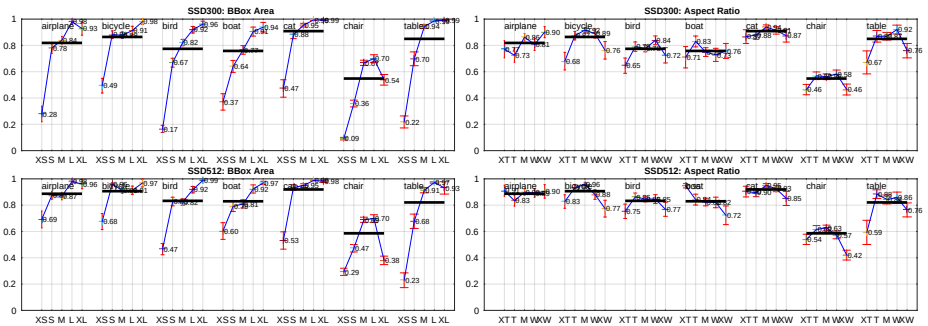


图4: 使用[21]方法在VOC2007测试集上不同物体特性的敏感性与影响。左图展示了各类别边界框面积的影响, 右图展示了宽高比的影响。关键说明: 边界框面积: XS=极小; S=小; M=中; L=大; XL=极大。宽高比: XT=极窄/细高; T=高; M=中等; W=宽; XW=极宽。

More default box shapes is better. As described in Sec. 2.2, by default we use 6 default boxes per location. If we remove the boxes with $\frac{1}{3}$ and 3 aspect ratios, the performance drops by 0.6%. By further removing the boxes with $\frac{1}{2}$ and 2 aspect ratios, the performance drops another 2.1%. Using a variety of default box shapes seems to make the task of predicting boxes easier for the network.

Atrous is faster. As described in Sec. 3, we used the atrous version of a subsampled VGG16, following DeepLab-LargeFOV [17]. If we use the full VGG16, keeping pool5 with $2 \times 2 - s2$ and not subsampling parameters from fc6 and fc7, and add conv5_3 for prediction, the result is about the same while the speed is about 20% slower.

Prediction source layers from:						mAP		# Boxes
conv4_3	conv7	conv8_2	conv9_2	conv10_2	conv11_2	use boundary boxes?		
						Yes	No	
✓	✓	✓	✓	✓	✓	74.3	63.4	8732
✓	✓	✓	✓	✓		74.6	63.1	8764
✓	✓	✓	✓			73.8	68.4	8942
✓	✓	✓				70.7	69.2	9864
✓	✓					64.2	64.4	9025
	✓					62.4	64.0	8664

Table 3: Effects of using multiple output layers.

Multiple output layers at different resolutions is better. A major contribution of SSD is using default boxes of different scales on different output layers. To measure the advantage gained, we progressively remove layers and compare results. For a fair comparison, every time we remove a layer, we adjust the default box tiling to keep the total number of boxes similar to the original (8732). This is done by stacking more scales of boxes on remaining layers and adjusting scales of boxes if needed. We do not exhaustively optimize the tiling for each setting. Table 3 shows a decrease in accuracy with fewer layers, dropping monotonically from 74.3 to 62.4. When we stack boxes of multiple scales on a layer, many are on the image boundary and need to be handled carefully. We tried the strategy used in Faster R-CNN [2], ignoring boxes which are on the boundary. We observe some interesting trends. For example, it hurts the performance by a large margin if we use very coarse feature maps (e.g. conv11_2 (1×1) or conv10_2 (3×3)). The reason might be that we do not have enough large boxes to cover large objects after the pruning. When we use primarily finer resolution maps, the performance starts increasing again because even after pruning a sufficient number of large boxes remains. If we only use conv7 for prediction, the performance is the worst, reinforcing the message that it is critical to spread boxes of different scales over different layers. Besides, since our predictions do not rely on ROI pooling as in [6], we do not have the *collapsing bins* problem in low-resolution feature maps [23]. The SSD architecture combines predictions from feature maps of various resolutions to achieve comparable accuracy to Faster R-CNN, while using lower resolution input images.

更多的默认框形状效果更好。如第2.2节所述，默认情况下我们每个位置使用6个默认框。如果移除具有 $\frac{1}{3}$ 和3种长宽比的框，性能会下降0.6%。若进一步移除具有 $\frac{1}{2}$ 和2种长宽比的框，性能会再下降2.1%。使用多种默认框形状似乎能让网络更轻松地预测边界框。

空洞卷积速度更快。如第3节所述，我们采用了DeepLab-LargeFOV[17]的方法，使用了经过子采样的VGG16的空洞卷积版本。若使用完整的VGG16，保持pool5为2×2步长且不从fc6和fc7层进行参数子采样，并添加conv5_3层进行预测，结果大致相同，但速度会降低约20%。

Prediction source layers from:						mAP		# Boxes
						use boundary boxes?		
conv4_3	conv7	conv8_2	conv9_2	conv10_2	conv11_2	Yes	No	
✓	✓	✓	✓	✓	✓	74.3	63.4	8732
✓	✓	✓	✓	✓		74.6	63.1	8764
✓	✓	✓	✓			73.8	68.4	8942
✓	✓	✓				70.7	69.2	9864
✓	✓					64.2	64.4	9025
	✓					62.4	64.0	8664

表3：使用多个输出层的影响。

在不同分辨率上设置多个输出层效果更佳。SSD的一个重要贡献是在不同输出层上使用不同尺度的默认框。为评估其优势，我们逐步移除层级并对比结果。为确保公平比较，每次移除层级时都会调整默认框的排布方式，使总框数接近原始值（8732个）。具体做法是在剩余层级上堆叠更多尺度的框，并在必要时调整框的尺度。我们并未对每种设置进行详尽的排布优化。表3显示随着层级减少，准确率从74.3单调下降至62.4。当在单层堆叠多尺度框时，大量边界框位于图像边缘需谨慎处理。我们尝试采用Faster R-CNN[2]的策略——忽略边缘位置的框，并观察到一些有趣现象：若使用过于粗糙的特征图（如conv11_2（1×1）或conv10_2（3×3）），性能会大幅下降，原因可能是剪枝后缺乏足够的大尺度框覆盖大型目标；当主要使用更精细的分辨率特征图时，由于剪枝后仍保留足量大尺度框，性能开始回升。仅使用conv7进行预测时效果最差，这印证了将不同尺度框分散到不同层级的关键性。此外，由于我们的预测不依赖[6]中的ROI池化，因此不存在低分辨率特征图的*collapsing bins*问题[23]。SSD架构通过融合多分辨率特征图的预测结果，在使用更低分辨率输入图像的同时达到了与Faster R-CNN相当的准确率。

3.3 PASCAL VOC2012

We use the same settings as those used for our basic VOC2007 experiments above, except that we use VOC2012 `trainval` and VOC2007 `trainval` and `test` (21503 images) for training, and test on VOC2012 `test` (10991 images). We train the models with 10^{-3} learning rate for 60k iterations, then 10^{-4} for 20k iterations. Table 4 shows the results of our SSD300 and SSD512⁴ model. We see the same performance trend as we observed on VOC2007 test. Our SSD300 improves accuracy over Fast/Faster R-CNN. By increasing the training and testing image size to 512×512 , we are 4.5% more accurate than Faster R-CNN. Compared to YOLO, SSD is significantly more accurate, likely due to the use of convolutional default boxes from multiple feature maps and our matching strategy during training. When fine-tuned from models trained on COCO, our SSD512 achieves 80.0% mAP, which is 4.1% higher than Faster R-CNN.

Method	data	mAP	aero	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv
Fast[6]	07++12	68.4	82.3	78.4	70.8	52.3	38.7	77.8	71.6	89.3	44.2	73.0	55.0	87.5	80.5	80.8	72.0	35.1	68.3	65.7	80.4	64.2
Faster[2]	07++12	70.4	84.9	79.8	74.3	53.9	49.8	77.5	75.9	88.5	45.6	77.1	55.3	86.9	81.7	80.9	79.6	40.1	72.6	60.9	81.2	61.5
Faster[2]	07++12+COCO	75.9	87.4	83.6	76.8	62.9	59.6	81.9	82.0	91.3	54.9	82.6	59.0	89.0	85.5	84.7	84.1	52.2	78.9	65.5	85.4	70.2
YOLO[5]	07++12	57.9	77.0	67.2	57.7	38.3	22.7	68.3	55.9	81.4	36.2	60.8	48.5	77.2	72.3	71.3	63.5	28.9	52.2	54.8	73.9	50.8
SSD300	07++12	72.4	85.6	80.1	70.5	57.6	46.2	79.4	76.1	89.2	53.0	77.0	60.8	87.0	83.1	82.3	79.4	45.9	75.9	69.5	81.9	67.5
SSD300	07++12+COCO	77.5	90.2	83.3	76.3	63.0	53.6	83.8	82.8	92.0	59.7	82.7	63.5	89.3	87.6	85.9	84.3	52.6	82.5	74.1	88.4	74.2
SSD512	07++12	74.9	87.4	82.3	75.8	59.0	52.6	81.7	81.5	90.0	55.4	79.0	59.8	88.4	84.3	84.7	83.3	50.2	78.0	66.3	86.3	72.0
SSD512	07++12+COCO	80.0	90.7	86.8	80.5	67.8	60.8	86.3	85.5	93.5	63.2	85.7	64.4	90.9	89.0	88.9	86.8	57.2	85.1	72.8	88.4	75.9

Table 4: **PASCAL VOC2012 test detection results.** Fast and Faster R-CNN use images with minimum dimension 600, while the image size for YOLO is 448×448 . data: "07++12": union of VOC2007 `trainval` and `test` and VOC2012 `trainval`. "07++12+COCO": first train on COCO `trainval`35k then fine-tune on 07++12.

3.4 COCO

To further validate the SSD framework, we trained our SSD300 and SSD512 architectures on the COCO dataset. Since objects in COCO tend to be smaller than PASCAL VOC, we use smaller default boxes for all layers. We follow the strategy mentioned in Sec. 2.2, but now our smallest default box has a scale of 0.15 instead of 0.2, and the scale of the default box on conv4_3 is 0.07 (e.g. 21 pixels for a 300×300 image)⁵.

We use the `trainval35k` [24] for training. We first train the model with 10^{-3} learning rate for 160k iterations, and then continue training for 40k iterations with 10^{-4} and 40k iterations with 10^{-5} . Table 5 shows the results on `test-dev2015`. Similar to what we observed on the PASCAL VOC dataset, SSD300 is better than Fast R-CNN in both mAP@0.5 and mAP@[0.5:0.95]. SSD300 has a similar mAP@0.75 as ION [24] and Faster R-CNN [25], but is worse in mAP@0.5. By increasing the image size to 512×512 , our SSD512 is better than Faster R-CNN [25] in both criteria. Interestingly, we observe that SSD512 is 5.3% better in mAP@0.75, but is only 1.2% better in mAP@0.5. We also observe that it has much better AP (4.8%) and AR (4.6%) for large objects, but has relatively less improvement in AP (1.3%) and AR (2.0%) for

⁴ <http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?cls=mean&challengeid=11&compid=4>

⁵ For SSD512 model, we add extra conv12_2 for prediction, set $s_{\min}$ to 0.1, and 0.04 on conv4_3.

3.3 PASCAL VOC2012

我们采用与上述基础VOC2007实验相同的设置，区别在于使用VOC2012 trainval与VOC2007 trainval及test数据集（共21503张图像）进行训练，并在VOC2012 test数据集（10991张图像）上测试。模型训练以 10^{-3} 学习率进行60k次迭代，随后以 10^{-4} 学习率进行20k次迭代。表4展示了我们的SSD300与SSD512⁴模型结果。我们观察到与VOC2007测试集一致的性能趋势：SSD300在准确率上超越了Fast/Faster R-CNN；当训练与测试图像尺寸提升至 512×512 时，模型比Faster R-CNN准确率高出4.5%。相较于YOLO，SSD凭借多特征图的卷积默认框及训练中的匹配策略实现了显著更高的精度。在基于COCO预训练模型进行微调后，我们的SSD512达到80.0% mAP，较Faster R-CNN提升4.1%。

方法 数据 mAP 航空器 自行车 鸟类 船只 瓶子 巴士 汽车 猫 椅子 牛 桌子 狗 马 摩托车 人 植物 羊 沙发 火车 电视 Fast[6] 07++12 68.4 82.3 78.4 70.8 52.3 38.7 77.8 71.6 89.3 44.2 73.0 55.0 87.5 80.5 80.8 72.0 35.1 68.3 65.7 80.4 64.2 Faster[2] 07++12 70.4 84.9 79.8 74.3 53.9 49.8 77.5 75.9 88.5 45.6 77.1 55.3 86.9 81.7 80.9 79.6 40.1 72.6 60.9 81.2 61.5 Faster[2] 07++12+COCO 75.9 87.4 83.6 76.8 62.9 59.6 81.9 82.0 91.3 54.9 82.6 59.0 89.0 85.5 84.7 84.1 52.2 78.9 65.5 85.4 70.2 YOLO[5] 07++12 57.9 77.0 67.2 57.7 38.3 22.7 68.3 55.9 81.4 36.2 60.8 48.5 77.2 72.3 71.3 63.5 28.9 52.2 54.8 73.9 50.8 SSD300 07++12 72.4 85.6 80.1 70.5 57.6 46.2 79.4 76.1 89.2 53.0 77.0 60.8 87.0 83.1 82.3 79.4 45.9 75.9 69.5 81.9 67.5 SSD300 07++12+COCO 77.5 90.2 83.3 76.3 63.0 53.6 83.8 82.8 92.0 59.7 82.7 63.5 89.3 87.6 85.9 84.3 52.6 82.5 74.1 88.4 74.2 SSD512 07++12 74.9 87.4 82.3 75.8 59.0 52.6 81.7 81.5 90.0 55.4 79.0 59.8 88.4 84.3 84.7 83.3 50.2 78.0 66.3 86.3 72.0 SSD512 07++12+COCO 80.0 90.7 86.8 80.5 67.8 60.8 86.3 85.5 93.5 63.2 85.7 64.4 90.9 89.0 88.9 86.8 57.2 85.1 72.8 88.4 75.9

表4：PASCAL VOC2012测试集检测结果。Fast与Faster R-CNN采用最小边长为600的图像，而YOLO的图像尺寸为 448×448 。数据说明：“07++12”表示VOC2007训练验证集、测试集与VOC2012训练验证集的并集；“07++12+COCO”指先在COCO trainval35k上训练，再在07++12数据集上微调。

3.4 COCO

为了进一步验证SSD框架，我们在COCO数据集上训练了SSD300和SSD512架构。由于COCO中的物体通常比PASCAL VOC中的更小，我们为所有层使用了更小的默认框。我们遵循第2.2节中提到的策略，但现在我们的最小默认框尺度为0.15而非0.2，并且conv4_3层上默认框的尺度为0.07（例如，对于 300×300 的图像对应21像素）⁵。

我们使用trainval35k [24]进行训练。首先以 10^{-3} 的学习率训练模型160k次迭代，然后继续以 10^{-4} 的学习率训练40k次迭代，再以 10^{-5} 的学习率训练40k次迭代。表5展示了在test-dev2015上的结果。与我们在PASCAL VOC数据集上观察到的类似，SSD300在mAP@0.5和mAP@[0.5:0.95]上均优于Fast R-CNN。SSD300在mAP@0.75上与ION [24]和Faster R-CNN [25]相近，但在mAP@0.5上表现较差。通过将图像尺寸增加到 512×512 ，我们的SSD512在两个指标上均优于Faster R-CNN [25]。有趣的是，我们观察到SSD512在mAP@0.75上提升了5.3%，但在mAP@0.5上仅提升了1.2%。我们还发现，对于大物体，其AP（4.8%）和AR（4.6%）有显著提升，但对于小物体，AP（1.3%）和AR（2.0%）的提升相对较小。

⁴ <http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?cls=mean&challengeid=11&compid=4>

⁵ For SSD512 model, we add extra conv12.2 for prediction, set $s_{\min}$ to 0.1, and 0.04 on conv4_3.

Method	data	Avg. Precision, IoU:			Avg. Precision, Area:			Avg. Recall, #Dets:			Avg. Recall, Area:		
		0.5:0.95	0.5	0.75	S	M	L	1	10	100	S	M	L
Fast [6]	train	19.7	35.9	-	-	-	-	-	-	-	-	-	-
Fast [24]	train	20.5	39.9	19.4	4.1	20.0	35.8	21.3	29.5	30.1	7.3	32.1	52.0
Faster [2]	trainval	21.9	42.7	-	-	-	-	-	-	-	-	-	-
ION [24]	train	23.6	43.2	23.6	6.4	24.1	38.3	23.2	32.7	33.5	10.1	37.7	53.6
Faster [25]	trainval	24.2	45.3	23.5	7.7	26.4	37.1	23.8	34.0	34.6	12.0	38.5	54.4
SSD300	trainval35k	23.2	41.2	23.4	5.3	23.2	39.6	22.5	33.2	35.3	9.6	37.6	56.5
SSD512	trainval35k	26.8	46.5	27.8	9.0	28.9	41.9	24.8	37.5	39.8	14.0	43.5	59.0

Table 5: **COCO test-dev2015 detection results.**

small objects. Compared to ION, the improvement in AR for large and small objects is more similar (5.4% vs. 3.9%). We conjecture that Faster R-CNN is more competitive on smaller objects with SSD because it performs two box refinement steps, in both the RPN part and in the Fast R-CNN part. In Fig. 5, we show some detection examples on COCO test-dev with the SSD512 model.

3.5 Preliminary ILSVRC results

We applied the same network architecture we used for COCO to the ILSVRC DET dataset [16]. We train a SSD300 model using the ILSVRC2014 DET train and val as used in [22]. We first train the model with 10^{-3} learning rate for 320k iterations, and then continue training for 80k iterations with 10^{-4} and 40k iterations with 10^{-5} . We can achieve 43.4 mAP on the val2 set [22]. Again, it validates that SSD is a general framework for high quality real-time detection.

3.6 Data Augmentation for Small Object Accuracy

Without a follow-up feature resampling step as in Faster R-CNN, the classification task for small objects is relatively hard for SSD, as demonstrated in our analysis (see Fig. 4). The data augmentation strategy described in Sec. 2.2 helps to improve the performance dramatically, especially on small datasets such as PASCAL VOC. The random crops generated by the strategy can be thought of as a “zoom in” operation and can generate many larger training examples. To implement a “zoom out” operation that creates more small training examples, we first randomly place an image on a canvas of $16\times$ of the original image size filled with mean values before we do any random crop operation. Because we have more training images by introducing this new “expansion” data augmentation trick, we have to double the training iterations. We have seen a consistent increase of 2%-3% mAP across multiple datasets, as shown in Table 6. In specific, Figure 6 shows that the new augmentation trick significantly improves the performance on small objects. This result underscores the importance of the data augmentation strategy for the final model accuracy.

An alternative way of improving SSD is to design a better tiling of default boxes so that its position and scale are better aligned with the receptive field of each position on a feature map. We leave this for future work.

Method	data	Avg. Precision, IoU:			Avg. Precision, Area:			Avg. Recall, #Dets:			Avg. Recall, Area:		
		0.5:0.95	0.5	0.75	S	M	L	1	10	100	S	M	L
Fast [6]	train	19.7	35.9	-	-	-	-	-	-	-	-	-	-
Fast [24]	train	20.5	39.9	19.4	4.1	20.0	35.8	21.3	29.5	30.1	7.3	32.1	52.0
Faster [2]	trainval	21.9	42.7	-	-	-	-	-	-	-	-	-	-
ION [24]	train	23.6	43.2	23.6	6.4	24.1	38.3	23.2	32.7	33.5	10.1	37.7	53.6
Faster [25]	trainval	24.2	45.3	23.5	7.7	26.4	37.1	23.8	34.0	34.6	12.0	38.5	54.4
SSD300	trainval35k	23.2	41.2	23.4	5.3	23.2	39.6	22.5	33.2	35.3	9.6	37.6	56.5
SSD512	trainval35k	26.8	46.5	27.8	9.0	28.9	41.9	24.8	37.5	39.8	14.0	43.5	59.0

表5: COCO test-dev2015检测结果。

小物体。与ION相比，大物体和小物体在平均召回率（AR）上的提升更为接近（5.4%对比3.9%）。我们推测，Faster R-CNN在较小物体上相对于SSD更具竞争力的原因在于其进行了两次边界框优化步骤——分别在区域提议网络（RPN）部分和Fast R-CNN部分。在图5中，我们展示了SSD512模型在COCO test-dev数据集上的一些检测示例。

3.5 ILSVRC初步结果

我们将用于COCO的相同网络架构应用于ILSVRC DET数据集[16]。我们使用ILSVRC2014 DET的训练集和val1集（如[22]所述）训练SSD300模型。首先以 10^{-3} 的学习率训练32万次迭代，随后以 10^{-4} 的学习率继续训练8万次迭代，最后以 10^{-5} 的学习率训练4万次迭代。在val2集[22]上我们获得了43.4%的mAP。这再次验证了SSD是一个适用于高质量实时检测的通用框架。

3.6 面向小目标精度的数据增强

与Faster R-CNN中后续的特征重采样步骤不同，SSD对小目标的分类任务相对困难，正如我们的分析所示（见图4）。第2.2节描述的数据增强策略显著提升了性能，尤其是在PASCAL VOC等小型数据集上。该策略生成的随机裁剪可视为一种“放大”操作，能产生许多更大的训练样本。为实现“缩小”操作以创建更多小训练样本，我们在进行任何随机裁剪前，会先将图像随机放置于填充均值的画布上，画布尺寸为原始图像的16×。由于通过引入这种新的“扩展”数据增强技巧增加了训练图像数量，我们必须将训练迭代次数加倍。如表6所示，我们在多个数据集中观察到mAP持续提升2%-3%。具体而言，图6表明新增强技巧显著提升了小目标检测性能。这一结果凸显了数据增强策略对最终模型精度的重要性。

改进SSD的另一种方法是设计更好的默认框平铺方式，使其位置和尺度与特征图上每个位置的感受野更精确地对齐。我们将这一问题留待未来研究。



Fig. 5: **Detection examples on COCO test-dev with SSD512 model.** We show detections with scores higher than 0.6. Each color corresponds to an object category.

Method	VOC2007 test		VOC2012 test		COCO test-dev2015		
	07+12	07+12+COCO	07++12	07++12+COCO	trainval35k		
	0.5	0.5	0.5	0.5	0.5:0.95	0.5	0.75
SSD300	74.3	79.6	72.4	77.5	23.2	41.2	23.4
SSD512	76.8	81.6	74.9	80.0	26.8	46.5	27.8
SSD300*	77.2	81.2	75.8	79.3	25.1	43.1	25.8
SSD512*	79.8	83.2	78.5	82.2	28.8	48.5	30.3

Table 6: **Results on multiple datasets when we add the image expansion data augmentation trick.** SSD300* and SSD512* are the models that are trained with the new data augmentation.



图5: 使用SSD512模型在COCO测试集上的检测示例。我们展示了得分高于0.6的检测结果。每种颜色对应一个物体类别。

Method	VOC2007 test		VOC2012 test		COCO test-dev2015		
	07+12	07+12+COCO	07++12	07++12+COCO	trainval35k		
	0.5	0.5	0.5	0.5	0.5:0.95	0.5	0.75
SSD300	74.3	79.6	72.4	77.5	23.2	41.2	23.4
SSD512	76.8	81.6	74.9	80.0	26.8	46.5	27.8
SSD300*	77.2	81.2	75.8	79.3	25.1	43.1	25.8
SSD512*	79.8	83.2	78.5	82.2	28.8	48.5	30.3

表6: 添加图像扩展数据增强技巧后在多个数据集上的结果。SSD300*和SSD512*是采用新数据增强方法训练的模型。

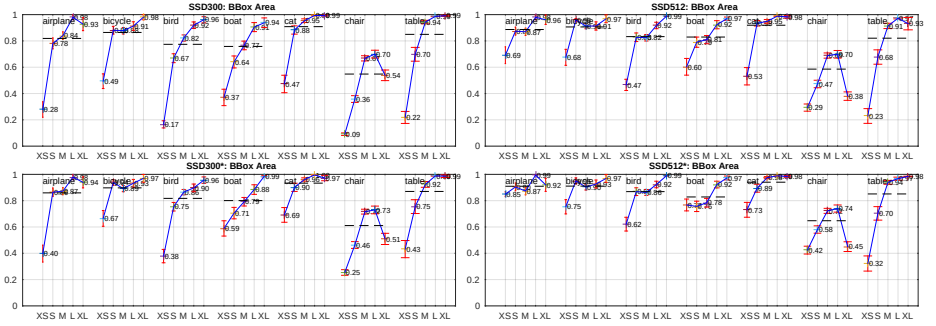


Fig. 6: **Sensitivity and impact of object size with new data augmentation on VOC2007 test set using [21].** The top row shows the effects of BBox Area per category for the original SSD300 and SSD512 model, and the bottom row corresponds to the SSD300* and SSD512* model trained with the new data augmentation trick. It is obvious that the new data augmentation trick helps detecting small objects significantly.

3.7 Inference time

Considering the large number of boxes generated from our method, it is essential to perform non-maximum suppression (nms) efficiently during inference. By using a confidence threshold of 0.01, we can filter out most boxes. We then apply nms with jaccard overlap of 0.45 per class and keep the top 200 detections per image. This step costs about 1.7 msec per image for SSD300 and 20 VOC classes, which is close to the total time (2.4 msec) spent on all newly added layers. We measure the speed with batch size 8 using Titan X and cuDNN v4 with Intel Xeon E5-2667v3@3.20GHz.

Table 7 shows the comparison between SSD, Faster R-CNN[2], and YOLO[5]. Both our SSD300 and SSD512 method outperforms Faster R-CNN in both speed and accuracy. Although Fast YOLO[5] can run at 155 FPS, it has lower accuracy by almost 22% mAP. To the best of our knowledge, SSD300 is the first real-time method to achieve above 70% mAP. Note that about 80% of the forward time is spent on the base network (VGG16 in our case). Therefore, using a faster base network could even further improve the speed, which can possibly make the SSD512 model real-time as well.

4 Related Work

There are two established classes of methods for object detection in images, one based on sliding windows and the other based on region proposal classification. Before the advent of convolutional neural networks, the state of the art for those two approaches – Deformable Part Model (DPM) [26] and Selective Search [1] – had comparable performance. However, after the dramatic improvement brought on by R-CNN [22], which combines selective search region proposals and convolutional network based post-classification, region proposal object detection methods became prevalent.

The original R-CNN approach has been improved in a variety of ways. The first set of approaches improve the quality and speed of post-classification, since it requires

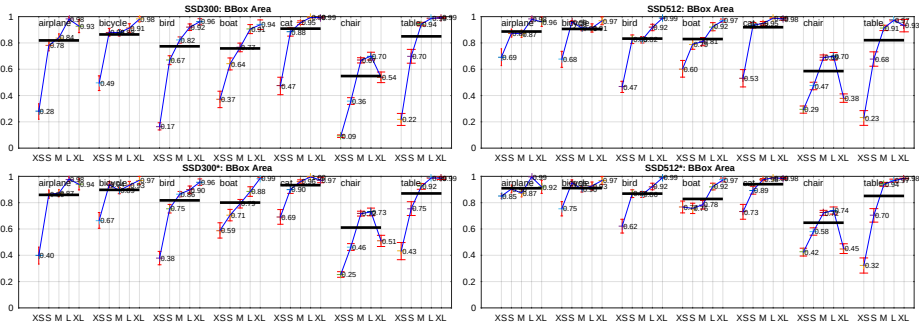


图6: 使用[21]在VOC2007测试集上, 新数据增强对物体尺寸的敏感性和影响。顶行显示了原始SSD300和SSD512模型在每个类别上的边界框面积(BBox Area)效果, 底行对应采用新数据增强技巧训练的SSD300*和SSD512*模型。显然, 新数据增强技巧显著提升了小物体的检测能力。

3.7 推理时间

考虑到我们方法生成的大量边界框, 在推理过程中高效执行非极大值抑制(nms)至关重要。通过使用0.01的置信度阈值, 我们可以过滤掉大部分边界框。随后对每个类别应用雅卡尔重叠率为0.45的nms处理, 并保留每张图像中置信度最高的200个检测结果。对于SSD300模型和20个VOC类别, 此步骤每张图像耗时约1.7毫秒, 接近所有新增层处理总时间(2.4毫秒)。我们使用Titan X显卡、cuDNN v4库及Intel Xeon E5-2667v3@3.20GHz处理器, 在批量大小为8的条件下测得该速度。

表7展示了SSD、Faster R-CNN[2]和YOLO[5]之间的比较。我们的SSD300和SSD512方法在速度和精度上都优于Faster R-CNN。尽管Fast YOLO[5]能以155 FPS运行, 但其准确率降低了近22% mAP。据我们所知, SSD300是首个实现超过70% mAP的实时方法。需要注意的是, 约80%的前向计算时间消耗在基础网络(本文中为VGG16)上。因此, 采用更快的基础网络可以进一步提升速度, 这可能使SSD512模型也能实现实时处理。

4 相关工作

图像中的目标检测方法主要有两大类: 一类基于滑动窗口, 另一类基于区域建议分类。在卷积神经网络出现之前, 这两种方法的代表技术——可变形部件模型(DPM) [26]和选择性搜索[1]——性能相当。然而, 随着R-CNN[22]将选择性搜索区域建议与基于卷积网络的后续分类相结合带来性能的显著提升, 基于区域建议的目标检测方法开始成为主流。

最初的R-CNN方法已在多个方面得到改进。第一类改进集中于提升后分类的质量与速度, 因其需要

Method	mAP	FPS	batch size	# Boxes	Input resolution
Faster R-CNN (VGG16)	73.2	7	1	~ 6000	$\sim 1000 \times 600$
Fast YOLO	52.7	155	1	98	448×448
YOLO (VGG16)	66.4	21	1	98	448×448
SSD300	74.3	46	1	8732	300×300
SSD512	76.8	19	1	24564	512×512
SSD300	74.3	59	8	8732	300×300
SSD512	76.8	22	8	24564	512×512

Table 7: **Results on Pascal VOC2007 test.** SSD300 is the only real-time detection method that can achieve above 70% mAP. By using a larger input image, SSD512 outperforms all methods on accuracy while maintaining a close to real-time speed.

the classification of thousands of image crops, which is expensive and time-consuming. SPPnet [9] speeds up the original R-CNN approach significantly. It introduces a spatial pyramid pooling layer that is more robust to region size and scale and allows the classification layers to reuse features computed over feature maps generated at several image resolutions. Fast R-CNN [6] extends SPPnet so that it can fine-tune all layers end-to-end by minimizing a loss for both confidences and bounding box regression, which was first introduced in MultiBox [7] for learning objectness.

The second set of approaches improve the quality of proposal generation using deep neural networks. In the most recent works like MultiBox [7,8], the Selective Search region proposals, which are based on low-level image features, are replaced by proposals generated directly from a separate deep neural network. This further improves the detection accuracy but results in a somewhat complex setup, requiring the training of two neural networks with a dependency between them. Faster R-CNN [2] replaces selective search proposals by ones learned from a region proposal network (RPN), and introduces a method to integrate the RPN with Fast R-CNN by alternating between fine-tuning shared convolutional layers and prediction layers for these two networks. This way region proposals are used to pool mid-level features and the final classification step is less expensive. Our SSD is very similar to the region proposal network (RPN) in Faster R-CNN in that we also use a fixed set of (default) boxes for prediction, similar to the anchor boxes in the RPN. But instead of using these to pool features and evaluate another classifier, we simultaneously produce a score for each object category in each box. Thus, our approach avoids the complication of merging RPN with Fast R-CNN and is easier to train, faster, and straightforward to integrate in other tasks.

Another set of methods, which are directly related to our approach, skip the proposal step altogether and predict bounding boxes and confidences for multiple categories directly. OverFeat [4], a deep version of the sliding window method, predicts a bounding box directly from each location of the topmost feature map after knowing the confidences of the underlying object categories. YOLO [5] uses the whole topmost feature map to predict both confidences for multiple categories and bounding boxes (which are shared for these categories). Our SSD method falls in this category because we do not have the proposal step but use the default boxes. However, our approach is more flexible than the existing methods because we can use default boxes of different aspect

Method	mAP	FPS	batch size	# Boxes	Input resolution
Faster R-CNN (VGG16)	73.2	7	1	~ 6000	~ 1000 × 600
Fast YOLO	52.7	155	1	98	448 × 448
YOLO (VGG16)	66.4	21	1	98	448 × 448
SSD300	74.3	46	1	8732	300 × 300
SSD512	76.8	19	1	24564	512 × 512
SSD300	74.3	59	8	8732	300 × 300
SSD512	76.8	22	8	24564	512 × 512

表7: Pascal VOC2007测试集上的结果。SSD300是唯一能达到70%以上mAP的实时检测方法。通过使用更大的输入图像, SSD512在保持接近实时速度的同时, 在准确率上超越了所有方法。

对数千个图像裁剪区域进行分类, 既昂贵又耗时。SPPnet [9] 显著加快了原始 R-CNN 方法的速度。它引入了空间金字塔池化层, 该层对区域大小和尺度更具鲁棒性, 并允许分类层复用在不同图像分辨率下生成的特征图中计算出的特征。Fast R-CNN [6] 扩展了 SPPnet, 使其能够通过同时最小化置信度和边界框回归的损失 (该方法最初由 MultiBox [7] 提出用于学习物体性) 来端到端地微调所有网络层。

第二类方法利用深度神经网络提升候选框生成的质量。在最新的研究如Multi Box[7,8]中, 基于底层图像特征的选择性搜索区域候选框被直接由独立深度神经网络生成的候选框所取代。这进一步提高了检测精度, 但也导致系统结构略显复杂, 需要训练两个相互依赖的神经网络。Faster R-CNN[2]通过区域候选网络 (RPN) 学习生成的候选框替代了选择性搜索候选框, 并提出一种交替微调共享卷积层与两个网络预测层的方法, 将RPN与Fast R-CNN相整合。这种方式利用区域候选框汇聚中层特征, 使最终分类步骤的计算成本降低。我们的SSD与Faster R-CNN中的区域候选网络 (RPN) 高度相似: 同样采用一组固定的 (默认) 边界框进行预测, 类似于RPN中的锚框。但我们并非用这些边界框汇聚特征后再交由其他分类器评估, 而是直接在每个边界框中同步生成每个物体类别的得分。因此, 我们的方法避免了RPN与Fast R-CNN融合的复杂性, 更易于训练、速度更快, 并能直接集成到其他任务中。

另一类方法完全跳过了候选框生成步骤, 直接预测多类别的边界框和置信度, 这与我们的方法直接相关。OverFeat [4] 作为滑动窗口方法的深度版本, 在获知底层物体类别的置信度后, 直接从顶层特征图的每个位置预测边界框。YOLO [5] 则利用整个顶层特征图同时预测多类别置信度及边界框 (这些边界框在各类别间共享)。我们的SSD方法属于此类, 因为我们无需候选框生成步骤, 而是采用默认框进行预测。然而, 我们的方法比现有方案更灵活, 因为我们可以使用不同宽高比的默认框。

ratios on each feature location from multiple feature maps at different scales. If we only use one default box per location from the topmost feature map, our SSD would have similar architecture to OverFeat [4]; if we use the whole topmost feature map and add a fully connected layer for predictions instead of our convolutional predictors, and do not explicitly consider multiple aspect ratios, we can approximately reproduce YOLO [5].

5 Conclusions

This paper introduces SSD, a fast single-shot object detector for multiple categories. A key feature of our model is the use of multi-scale convolutional bounding box outputs attached to multiple feature maps at the top of the network. This representation allows us to efficiently model the space of possible box shapes. We experimentally validate that given appropriate training strategies, a larger number of carefully chosen default bounding boxes results in improved performance. We build SSD models with at least an order of magnitude more box predictions sampling location, scale, and aspect ratio, than existing methods [5,7]. We demonstrate that given the same VGG-16 base architecture, SSD compares favorably to its state-of-the-art object detector counterparts in terms of both accuracy and speed. Our SSD512 model significantly outperforms the state-of-the-art Faster R-CNN [2] in terms of accuracy on PASCAL VOC and COCO, while being $3\times$ faster. Our real time SSD300 model runs at 59 FPS, which is faster than the current real time YOLO [5] alternative, while producing markedly superior detection accuracy.

Apart from its standalone utility, we believe that our monolithic and relatively simple SSD model provides a useful building block for larger systems that employ an object detection component. A promising future direction is to explore its use as part of a system using recurrent neural networks to detect and track objects in video simultaneously.

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在不同尺度的多个特征图上，每个特征位置的长宽比。如果我们仅使用最顶层特征图上每个位置的一个默认框，我们的SSD将具有与OverFeat [4] 类似的架构；如果我们使用整个最顶层特征图并添加一个全连接层进行预测，而不是使用我们的卷积预测器，并且不显式考虑多种长宽比，我们可以近似复现YOLO [5]。

5 结论

本文介绍了SSD，一种用于多类别检测的快速单次目标检测器。我们模型的一个关键特征是在网络顶部的多个特征图上使用多尺度卷积边界框输出。这种表示方法使我们能够高效地建模可能的边界框形状空间。我们通过实验验证，在适当的训练策略下，使用大量精心选择的默认边界框可以提高性能。我们构建的SSD模型在采样位置、尺度和宽高比上的边界框预测数量，比现有方法[5,7]至少多一个数量级。我们证明，在相同的VGG-16基础架构下，SSD在准确性和速度方面均优于当前最先进的目标检测器。我们的SSD512模型在PASCAL VOC和COCO数据集上的准确性显著优于当前最优的Faster R-CNN[2]，同时速度提高了3×倍。我们的实时SSD300模型以59 FPS的速度运行，比当前的实时YOLO[5]方案更快，同时检测精度显著更高。

除了其独立的实用性之外，我们相信我们单一且相对简单的SSD模型为采用物体检测组件的更大系统提供了一个有用的构建模块。一个前景广阔的未来方向是探索将其作为系统的一部分，该系统使用循环神经网络来同时检测和跟踪视频中的物体。

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